Secret-Key PIR from One-Way Functions

Nir Bitansky & Noam Mazor

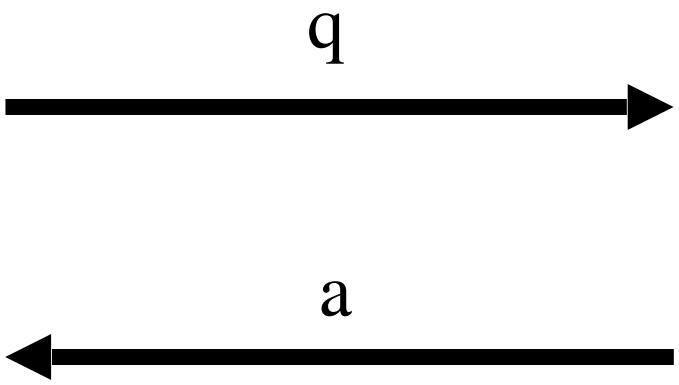
NYU

Private Information Retrieval

Chor–Goldreich–Kushilevitz–Sudan ’95/’98, Kushilevitz–Ostrovsky ’97

Client

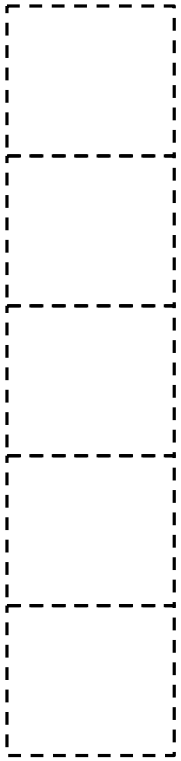
$i \in [N]$



D[i]

Server

$D \in \{0,1\}^N$



Goal: Hide i, $|q| + |a| \ll N$

Secret-Key PIR

Beimel–Ishai–Malkin ’04, Boyle–Ishai–Pass–Wootters ’17, Canetti–Holmgren–Richelson ’17

Client

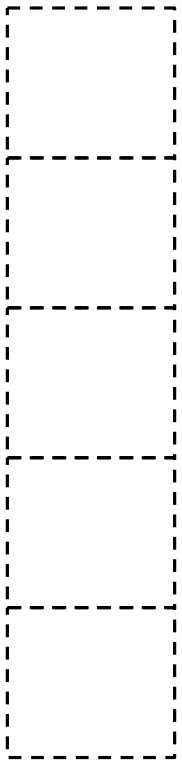
Offline

Server

D

sk

Preprocess(sk,D)



Secret-Key PIR

Beimel–Ishai–Malkin ’04, Boyle–Ishai–Pass–Wootters ’17, Canetti–Holmgren–Richelson ’17

Client

Offline

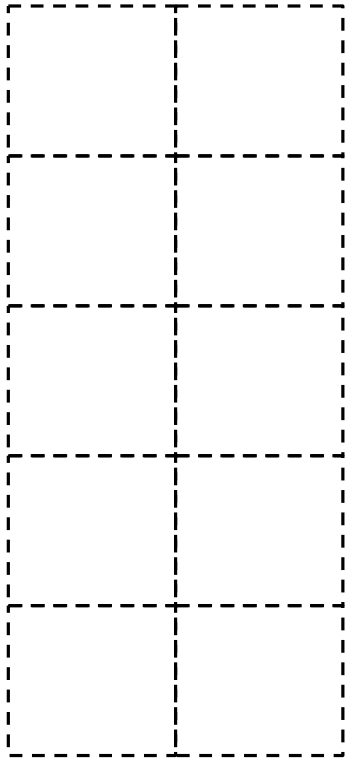
Server

sk

Preprocess(sk,D)



$\hat{D}$



Secret-Key PIR

Beimel–Ishai–Malkin ’04, Boyle–Ishai–Pass–Wootters ’17, Canetti–Holmgren–Richelson ’17

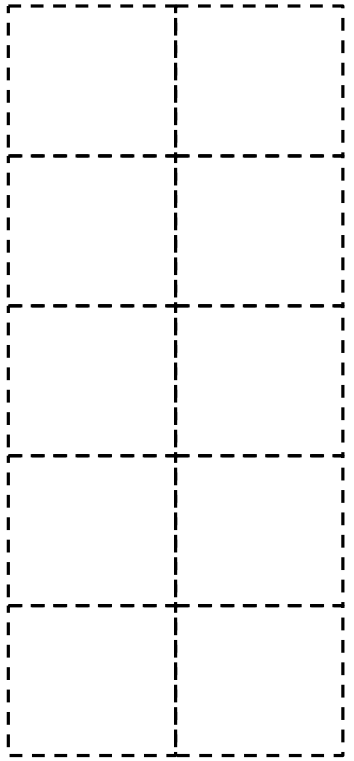
Client

Online

Server

sk

$\hat{D}$



Secret-Key PIR

Client

sk,i

D[i]

Online

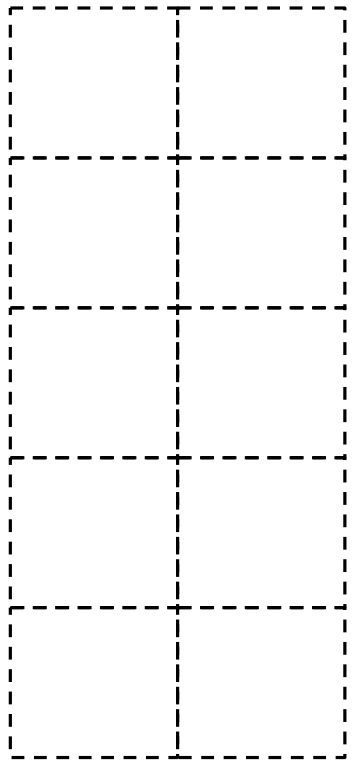
$q(sk, i)$

$a(\hat{D}, q)$

Same sk. Many queries.
Stateless

Server

$\hat{D}$



Why SK-PIR?

Double online efficiency

Beimel–Ishai–Malkin ’04, Boyle–Ishai–Pass–Wootters ’17, Canetti–Holmgren–Richelson ’17

By now achievable with public preprocessing under RLWE.

Lin–Mook–Wichs ’23

Other efficiency measures

Storage, preprocessing time, …

Secret-key cryptography

(plain) PIR → public-key cryptography

DiCrescenzo-Malkin-Ostrvosky ’00

Focus: communication efficiency

Secret-Key PIR from High-Noise LPN

Chen–Ishai–Mour–Rosen ’26

$$N^{\varepsilon}$$

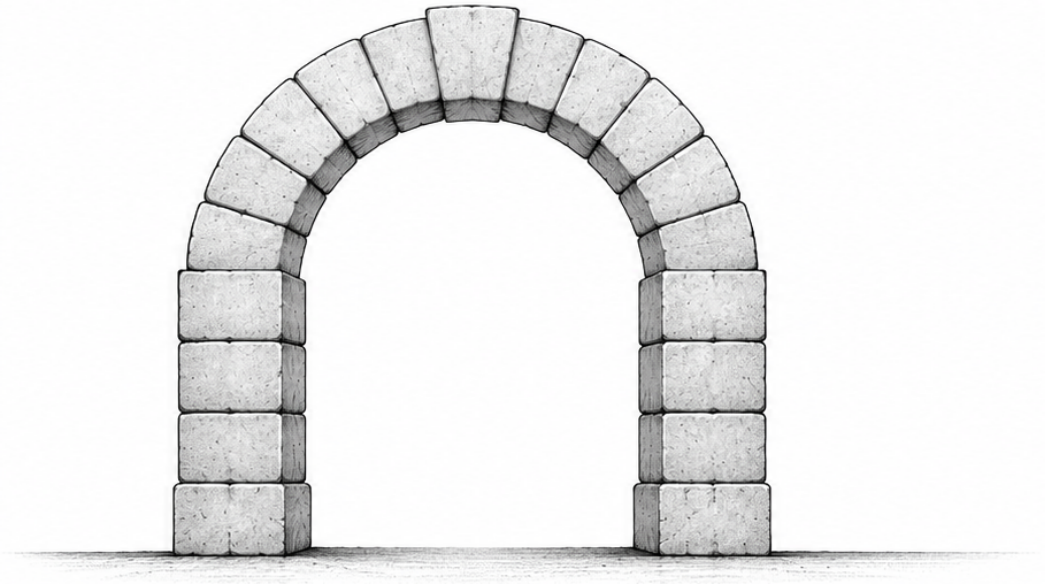
communication for any constant $\varepsilon > 0$

LPN with noise $n^{-0.01} \gg n^{-0.5}$

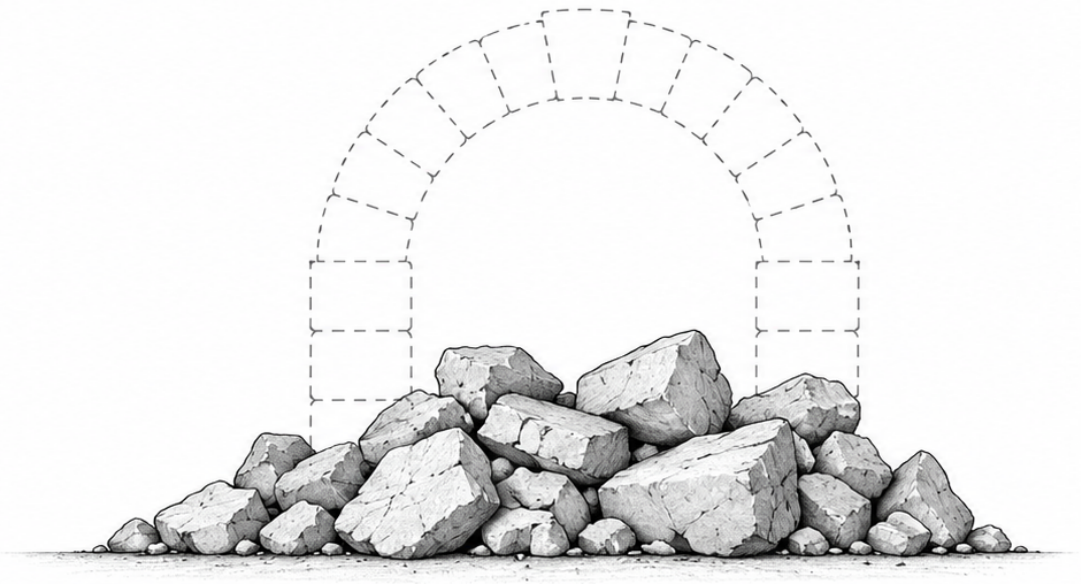
Not known to imply public-key crypto

Exploits LPN’s linear structure…

Is Structure Inherent in SK-PIR?



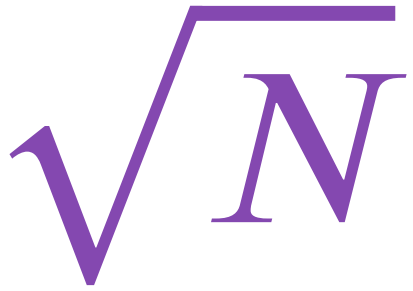
LPN



One-way functions?

Random functions?

Our Result: One-Way Functions (Partially) Suffice



online communication

up to security parameter and polylogarithmic factors

Rest of this talk: the construction

Overview

1. Construct SK-PIR with:

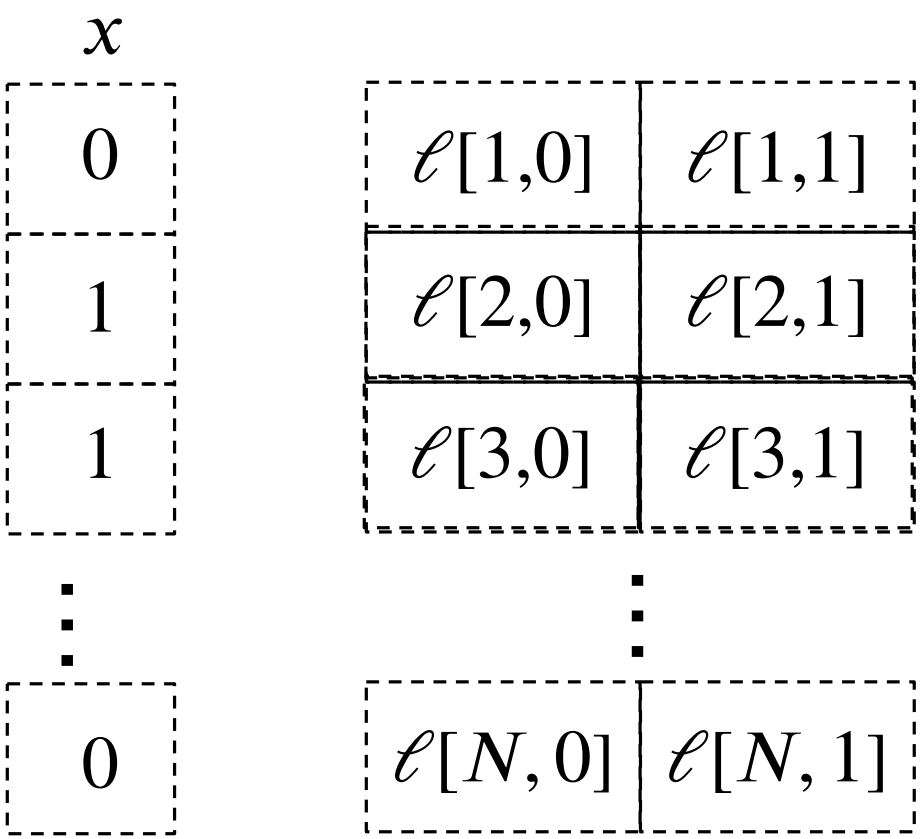
    - Long client message (N bits), but
    - Short server message (1 bit)

2. Balance communication

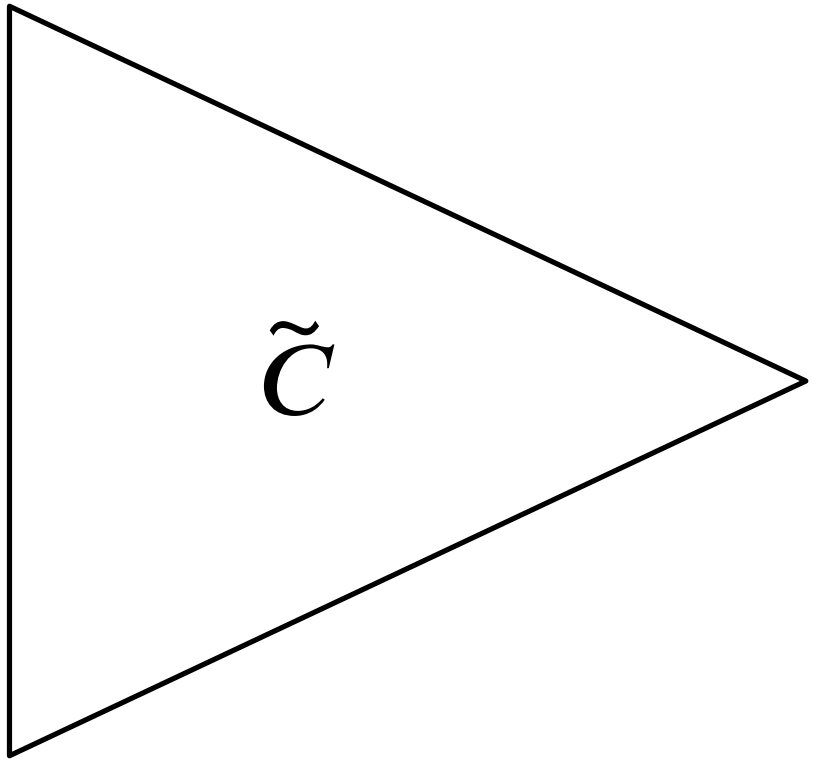
Recalling Garbled Circuits

Yao ’86

$$\text{Garble}(C: \{0,1\}^N \to \{0,1\})$$



Input labels

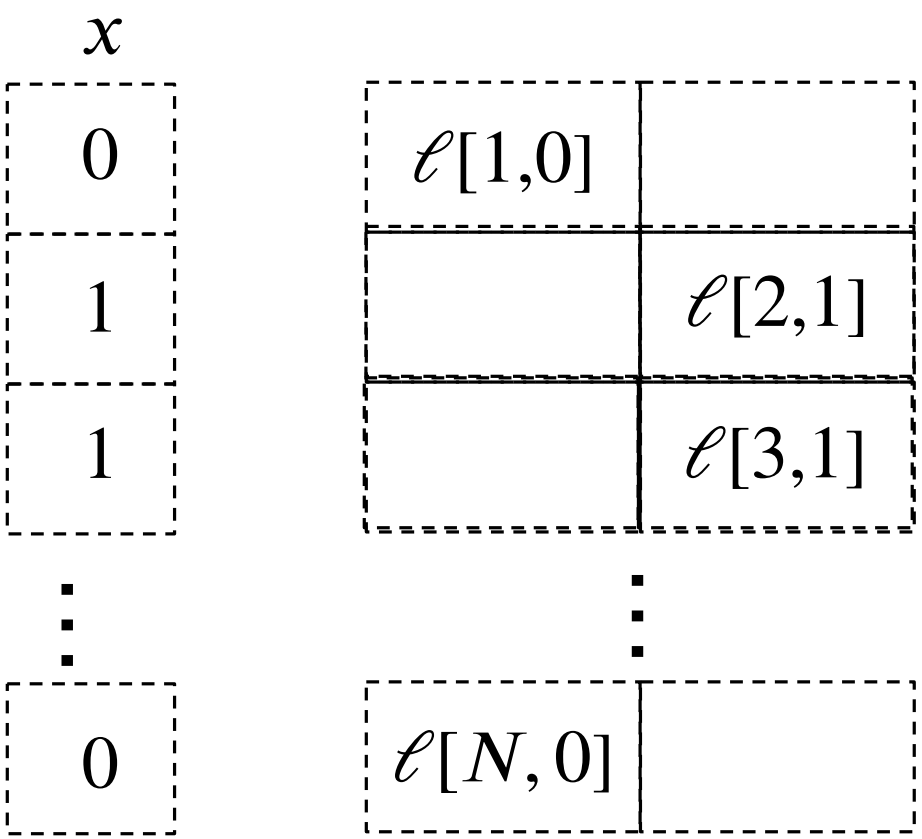


Garbled circuit, $|C| \approx |\tilde{C}|$

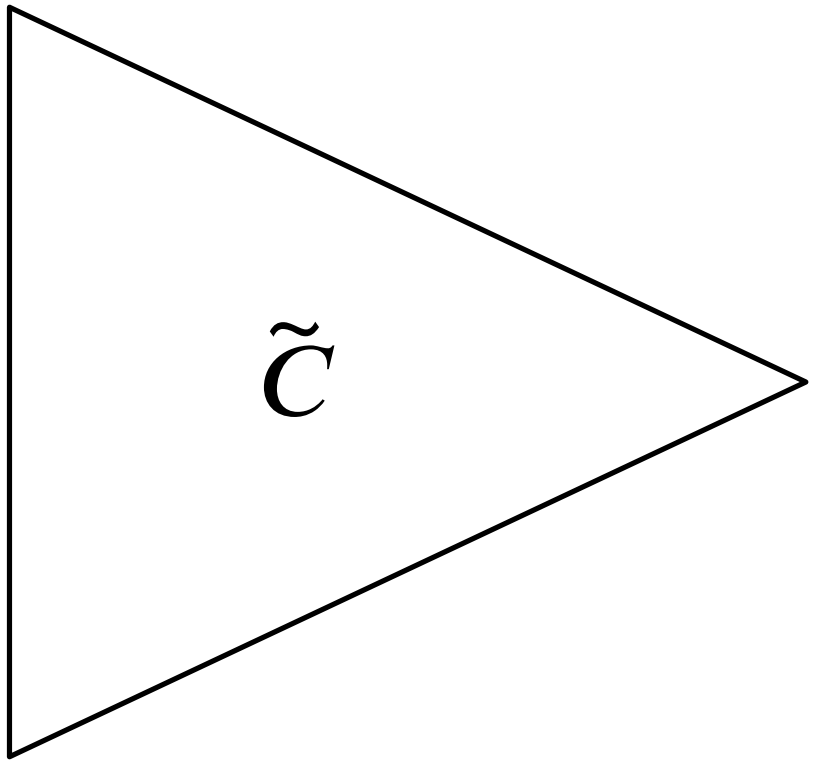
Recalling Garbled Circuits

Yao '86

$$\text{Garble}(sk, C: \{0,1\}^N \rightarrow \{0,1\})$$



Input labels

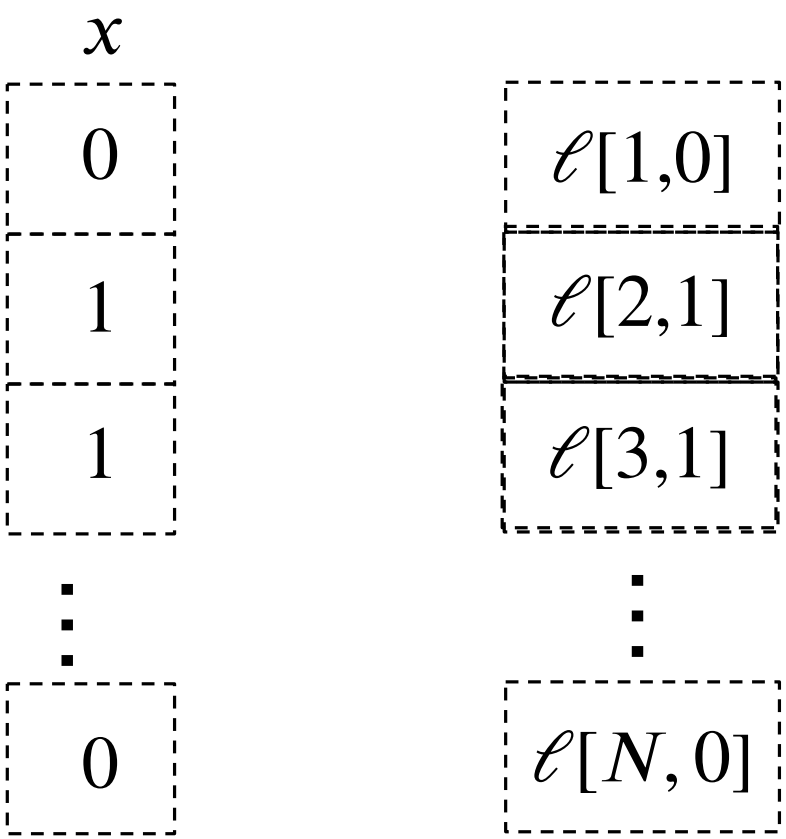


Garbled circuit, $|C| \approx |\tilde{C}|$

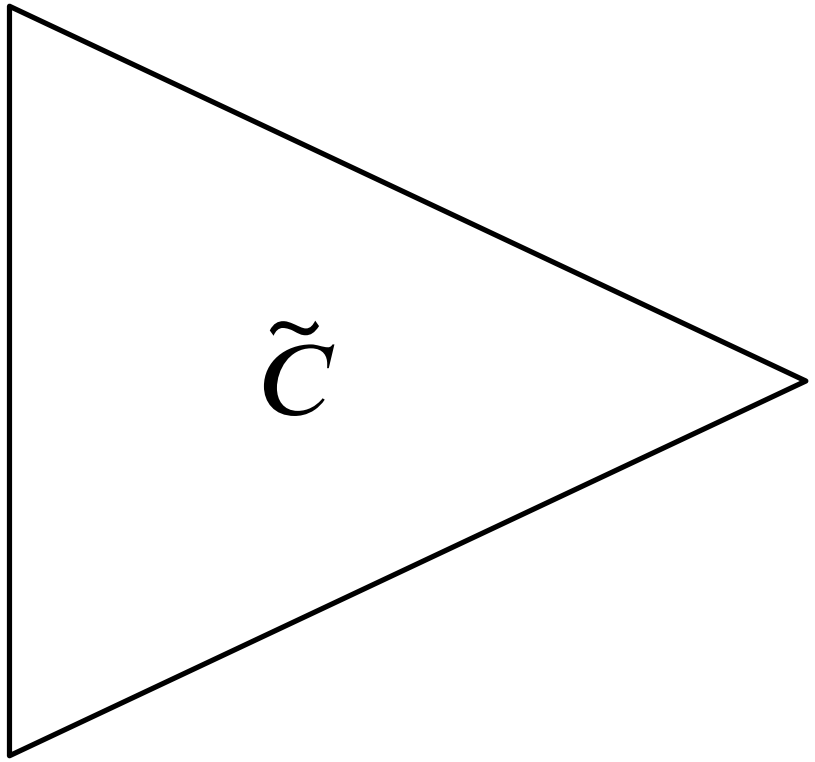
Recalling Garbled Circuits

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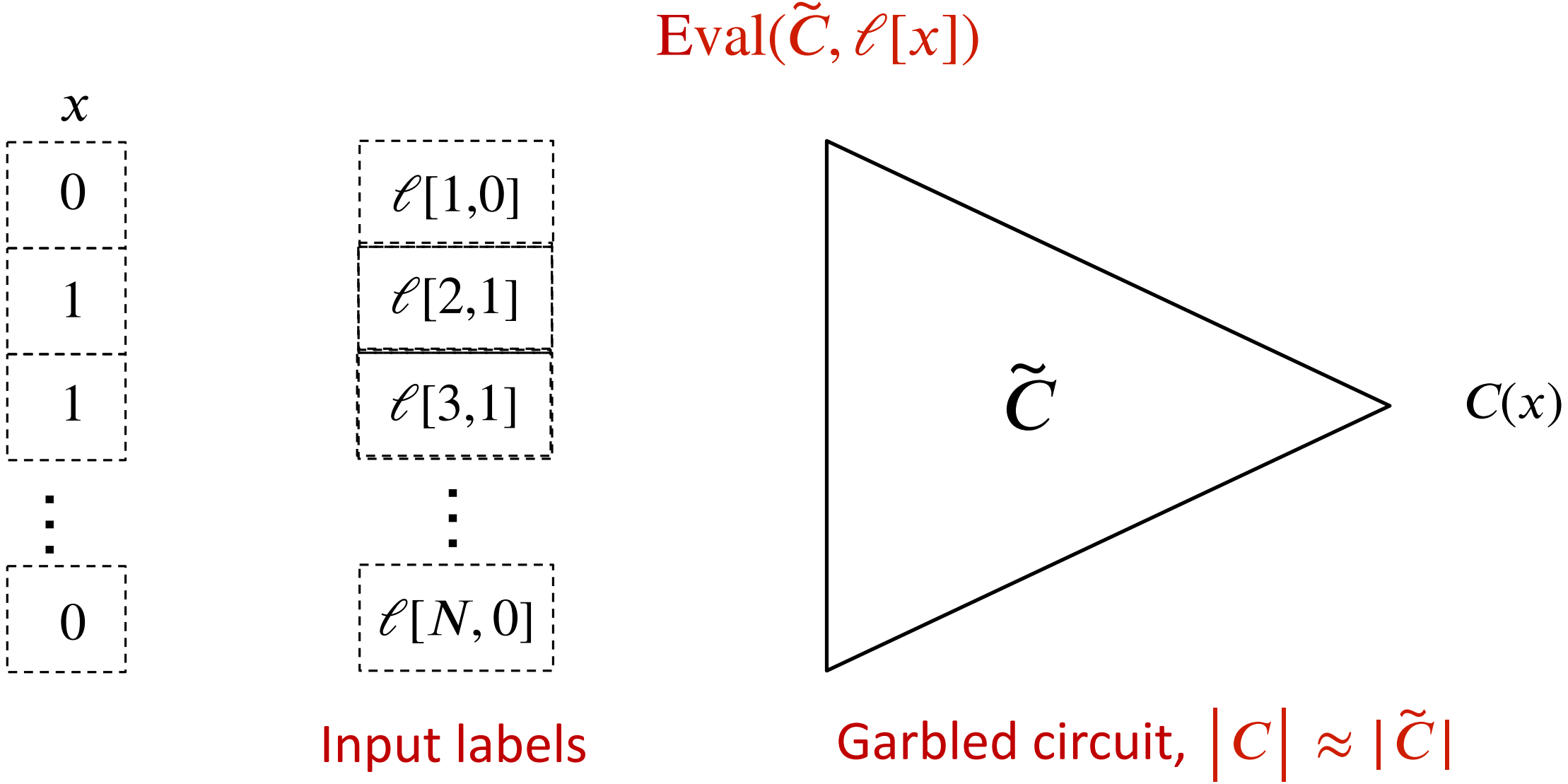
Input labels



Garbled circuit, $|C| \approx |\tilde{C}|$

Recalling Garbled Circuits

Yao ’86



The SK-PIR Preprocessing

Client

sk

$K[1,0]$	$K[1,1]$
$K[2,0]$	$K[2,1]$
$K[3,0]$	$K[3,1]$
⋮	
$K[N,0]$	$K[N,1]$

Server

D	$\hat{D}$	
0	$K[1,0]$	
1		$K[2,1]$
1		$K[3,1]$
⋮	⋮	
0	$K[N,0]$	

Long Query, Short Answer

Inspired by functional encryption [Sahai–Seyalioglu 2010]

Client

i, sk

$K[1,0]$	$K[1,1]$
$K[N,0]$	$K[N,1]$

$C_i(D) = D[i]$

$\ell[1,0]$	$\ell[1,1]$
$\ell[2,0]$	$\ell[2,1]$
$\ell[3,0]$	$\ell[3,1]$
⋮	
$\ell[N,0]$	$\ell[N,1]$

Input labels

$\tilde{C}_i$

Server

$\hat{D}$

$D[1]$	$K[1,D[1]]$
$D[N]$	$K[N,D[N]]$

Long Query, Short Answer

Inspired by functional encryption [Sahai–Seyalioglu 2010]

Client

i, sk

$K[1,0]$	$K[1,1]$
$K[N,0]$	$K[N,1]$

$C_i(D) = D[i]$

$\tilde{C}_i$

$\left\{ Enc_{K[j,b]}\left(\ell[j,b]\right) \right\}_{j,b}$

$\tilde{C}_i\left(\ell[D]\right) = D[i]$

Server

$\hat{D}$

$D[1]$	$K[1,D[1]]$
$D[N]$	$K[N,D[N]]$

Long Query, Short Answer

Inspired by functional encryption [Sahai–Seyalioglu 2010]

Client

i, sk

$K[1,0]$	$K[1,1]$
$K[N,0]$	$K[N,1]$

$C_{i,r}(D) = D[i] \oplus r$

$\tilde{C}_{i,r}$

$\left\{ Enc_{K[j,b]}\left(\ell[j,b]\right) \right\}_{j,b}$

$\approx N$

$\tilde{C}_{i,r}\left(\ell[D]\right) = D[i] \oplus r$

1

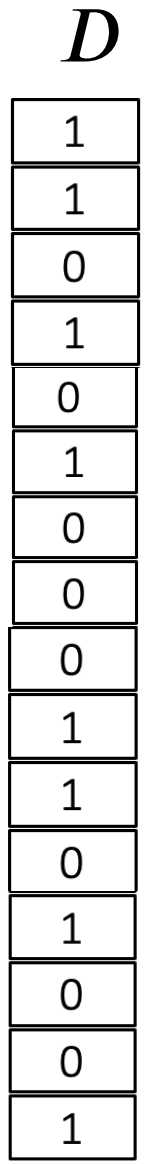
Server

$\hat{D}$

$D[1]$	$K[1,D[1]]$
$D[N]$	$K[N,D[N]]$

Balancing Communication

Kushilevitz-Ostrovsky ’97



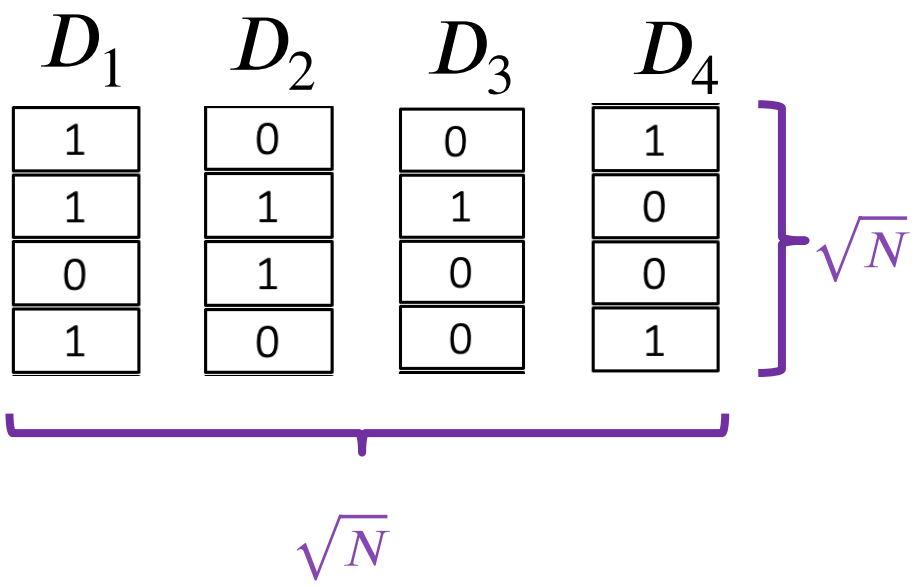
Balancing Communication

Kushilevitz-Ostrovsky '97

Client

$sk_1, \ldots, sk_{\sqrt{N}}$

Server



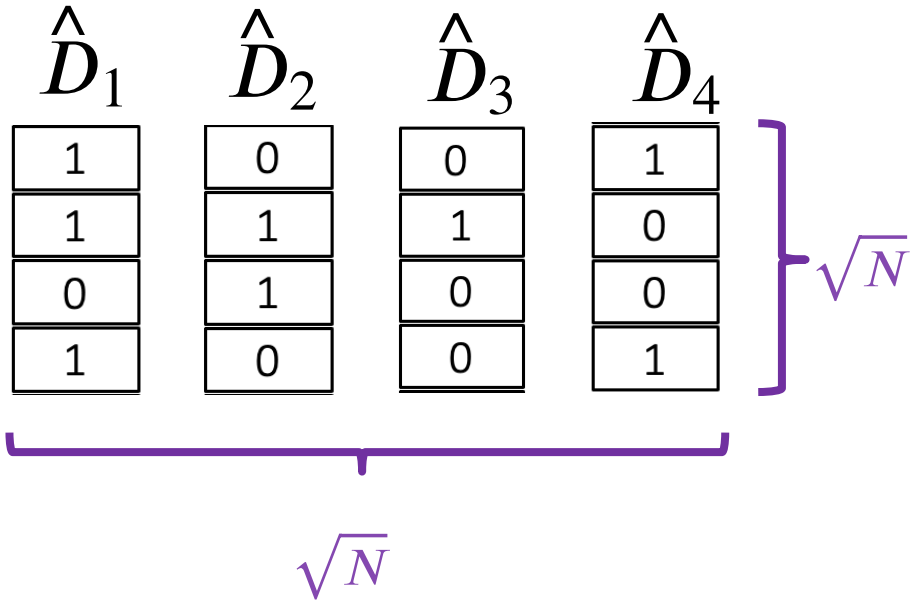
Balancing Communication

Kushilevitz-Ostrovsky '97

Client

$sk_1, \ldots, sk_{\sqrt{N}}$

Server



Balancing

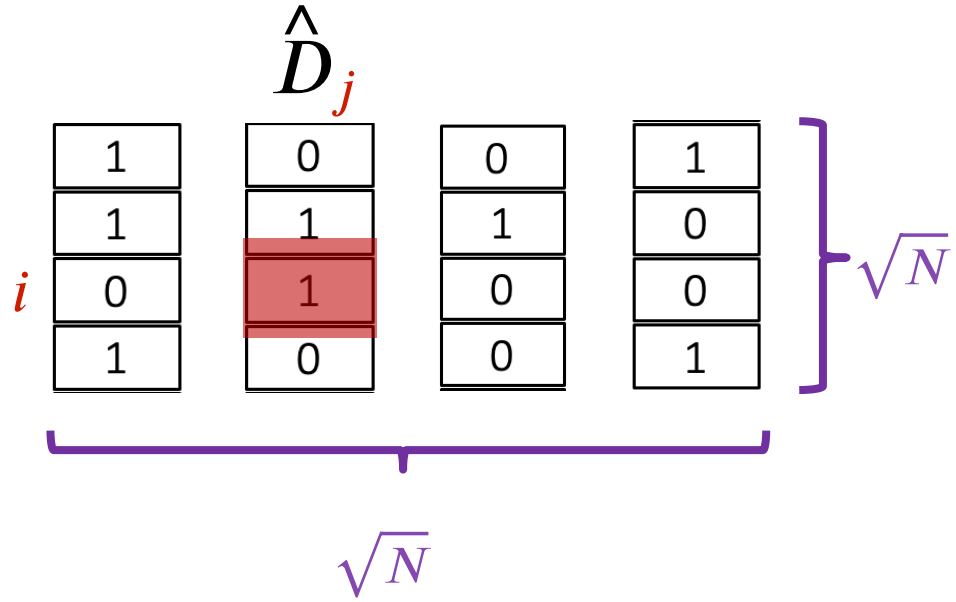
Kushilevitz-Ostrovsky '97

Client

$sk_1, \ldots, sk_{\sqrt{N}}$

(i, j)

Server



$\mathrm{q}(\mathrm{sk}_j, i)$

$\mathrm{a}\left(\hat{\mathrm{D}}_1, \mathrm{q}\right), \ldots, \mathrm{a}(\hat{\mathrm{D}}_{\sqrt{N}}, \mathrm{q})$

Communication: $\approx \sqrt{N}$

Security?

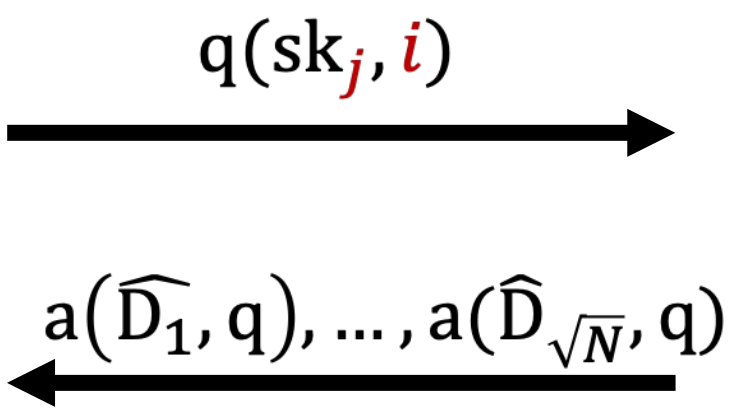
Security?

Can $\mathrm{q}(\mathrm{sk}_j, \mathrm{i})$ leak j?

- It might
- No, when:
  - Real labels look like random independent labels

"point&permute": Beaver–Micali–Rogaway '90; Brakerski–Lombardi–Segev–Vaikuntanathan '18

  - Enc(random) is random independent of key



Summary

Secret-Key PIR with communication $\sqrt{N}$ from one-way functions.

Open Question:

- Less communication?
- [Bitansky-Couteau-M]: Interactive SK-PIR with polylog communication in the ROM

Thanks!